

integrate several small producers of electricity, including wind and solar power. This is done using advanced power electronics that help to combine DC storage with renewable energy resources, by mitigating the variability.

Transmission too is made smarter in a modern grid using a wide network of sensors and control tools that avoid, detect and overcome faults. Advanced power electronics are being used to ensure normal voltage at all times, increase line capacity by series compensation and ride through smoothly when faults occur or large motors start.

The distribution of power to substations is perhaps the area of the grid most impacted by smart technologies. By studying consumption patterns and planning distribution accordingly, it is possible to manage demand-supply vagaries. There are also several technologies developed for this crucial segment of the smart grid. One recent development, for example, is conservation voltage reduction (CVR), a set of technologies that lowers voltages on distribution circuits to save energy.

At the retail level, utilities use automated metering infrastructure (AMI) with smart meters that help customers to understand their own consumption patterns and empower them to alter it positively using schemes like happy-hour discounts.

A job for electronics and communications...

As you would have surmised by now, there is a lot of work for electronics in a smart grid, from sensors, reclosers and phasor measurement units (PMUs), to data concentrators and power line communication (PLC) systems, wireless chips and smart meters with bidirectional communications. Most of these electronics are 'smart' in the sense that they can communicate with each other and a central server through a robust communication backbone, thereby enabling collection of data, their analysis and actions based on that. The backbone of all this smartness is therefore 'big data' sitting on the Cloud and integrated software solu-

Sense, respond and heal

A smart grid is smart because it can sense, respond and heal. Here is how it does so:

1. A smart grid includes several sensors and embedded instrumentation systems that monitor loads and sources of electricity constantly at varied locations.
2. The embedded systems transmit this information to centralised servers using wired or wireless networks.
3. Constant analysis of measurement data based on historical information and adaptive algorithms helps the grid to react intelligently to different conditions.
4. Intelligent distributed systems with real-time communications and embedded analytics take timely action based on this data, to ensure optimal performance and least downtime.
5. The smart grid is also self-healing; it isolates problems before they become too big, reorganises the grid and reroutes transmission so that things look normal to consumers till the problem is fixed by technicians.
6. "For example, during a thunderstorm, a flicker of light is because of a recloser in action. A smart recloser combines electrical switchgear to interrupt fault currents and real-time signal processing to monitor power signals and determine when to open the switch. The devices work like a smart circuit breaker because they protect the circuit in case of fault currents, but they also can 'reclose' to restore service and prevent downtime," says Satish Mohanram, technical marketing manager, National Instruments, India.

The physical transformation

1. A smart grid will be made of several micro-grids and one mammoth smart high-voltage grid that acts as the backbone.
2. The process of smartening the grid starts by adding processors to the components that carry electricity from the generators, including switches, circuit breakers, transformers and busbars.
3. This will enable the transmission lines to communicate to each other and monitor their individual pieces of the grid.
4. Electromechanical switches have to be replaced with solid-state, power-electronic circuits that can handle very high transmission voltages of 345kV and beyond.
5. This upgrade from analogue to secure digital will allow the entire network to be digitally controlled, making the smart grid's key functions of real-time self-monitoring and self-healing possible.

Based on <http://smartgrid.ieee.org/march-2013/813-the-self-healing-grid-a-concept-two-decades-in-the-making>

A multi-billion dollar market

The smart grid market is expected to cross \$400 billion worldwide by 2020, according to a recent study by GTM Research. Naturally, the scene is buzzing with players across a broad spectrum, offering products ranging from chips and switches to smart meters and IT solutions. The opportunities lie in advanced metering infrastructure (AMI), grid networking, transmission and distribution automation, distributed generation (DG) integration, renewable integration, storage, demand-response, home-area networking (of energy-efficient smart devices), building automation, security, Big Data, integrated smart grid solutions and consultancy, etc.

tions that support decision-making.

Grid infrastructure. There are components from data concentrators to PLC systems and wireless chips embedded in grid infrastructure. The data concentrators, for example, help to aggregate data from a certain number of meters and send them to the utility servers. These concentrators would generally need some wireless communication capability to acquire data and communication from the

concentrator to the utility servers via Ethernet, GSM, GPRS, WiMAX or telecom networks.

Similarly, sub-station control is also an electronics-intensive job today. Microprocessor-based relays are used for online monitoring and control replacing older SCADA systems; traditional switchboards with panels, switches and lights have given way to graphical user interfaces that can be accessed through PCs. Sundry sen-